

The number of unpaired electrons in cobalt atom is (atomic number of Co = 27)

- (a) 2
- (b) 3**
- (c) 4
- (d) 1

Answer: (b) 3

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

82. Zn is related to which group

- (a) IIB**
- (b) IIA
- (c) IA
- (d) IB

Answer: (a) IIB

Explanation: The correct choice is IIB. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

83. Which of the following element does not show variable valency

- (a) Ni
- (b) Zn**
- (c) Cu
- (d) Mn

Answer: (b) Zn

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

84. Which of the following is diamagnetic transitional metal ion

- (a) Ni+2
- (b) Zn+2**
- (c) Co+2
- (d) Cu+2

Answer: (b) Zn+2

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

85. Which of the following is not an actinide

- (a) Curium
- (b) Californium
- (c) Uranium
- (d) Terbium**

Answer: (d) Terbium

Explanation: Actinoids involve 5f orbitals; the small energy differences among 5f, 6d and 7s levels allow several oxidation states and extensive complex formation.



86. The ability to form complex compounds by the transitional metal ion is due to

- (a) Small size
- (b) Vacant 'd' orbitals
- (c) High nuclear charge
- (d) All of these**

Answer: (d) All of these

Explanation: Transition-metal ions form complexes readily because of their relatively high charge density and available orbitals for accepting ligand electron pairs.

87. Which of the following will give green hydrated ion

- (a) and
- (b) both
- (c)**
- (d) V+3

Answer: (c)

Explanation: The correct choice is . This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

88. Magnetic moment is expressed in

- (a) Faraday
- (b) Calorie
- (c) Bohr Magneton**
- (d) Debye

Answer: (c) Bohr Magneton

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

89. If the colours of salts of transition elements are due to the presence of unpaired electrons in the transition metal ions, which of the following ions will be colourless in aqueous solution

- (a) Ti3+
- (b) Ti4+**
- (c) Fe2+
- (d) Fe3+ (The at. no. of Ti and Fe are 22 and 26 respectively)

Answer: (b) Ti4+

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

90. In the periodic table first transition series is placed in

- (a) Third period
- (b) Fourth period**
- (c) Fifth period
- (d) None of these

Answer: (b) Fourth period

Explanation: The correct choice is Fourth period. This follows from the standard periodic and electronic properties of d-



and f-block elements tested in this question.

91. The element having general electronic configuration $3d^4 4s^1$ is

- (a) Noble gas
- (b) Non-metal
- (c) Metalloid
- (d) Transition metal**

Answer: (d) Transition metal

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

92. Which of the following general configuration of outermost shell represents chromium element [Cr's atomic number = 24]

- (a) $d^5 s^1$**
- (b) $d^6 s^0$
- (c) $d^4 s^2$
- (d) $d^3 s^2$

Answer: (a) $d^5 s^1$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

93. Which element gives maximum balanced oxide

- (a) V

- (b) Cr
- (c) Mn**
- (d) Fe

Answer: (c) Mn

Explanation: The correct choice is Mn. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

94. In first transition series, the melting point of Mn is low because

- (a) Due to d^{10} configuration, metallic bonds are strong
- (b) Due to d^7 configuration, metallic bonds are weak
- (c) Due to d^5 configuration, metallic bonds are weak**
- (d) None of these

Answer: (c) Due to d^5 configuration, metallic bonds are weak

Explanation: Melting/boiling trends depend largely on the strength of metallic bonding, which is influenced by the number of valence d and s electrons available for bonding.

95. Which of the following ions has the least magnetic moment

- (a) Cu^{+2}**
- (b) Ni^{+2}
- (c) Co^{+3}
- (d) Fe^{+2}

Answer: (a) Cu^{+2}



Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

Explanation: The correct choice is They crystallize with body centred cubic and hexagonal close-packed structures only. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

96. Among the following outermost configurations of transition metals, which shows the highest oxidation state

- (a) 3d34s2
- (b) 3d54s1
- (c) 3d54s2**
- (d) 3d64s2

Answer: (c) 3d54s2

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

98. The most malleable of all the metals is

- (a) Silver
- (b) Sodium
- (c) Gold**
- (d) Platinum

Answer: (c) Gold

Explanation: The correct choice is Gold. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

97. Which of the following is not true for transition metals

- (a) They are malleable and ductile
- (b) They have high boiling and melting points
- (c) They crystallize with body centred cubic and hexagonal close-packed structures only**
- (d) They show variable oxidation states although not always

Answer: (c) They crystallize with body centred cubic and hexagonal close-packed structures only

99. Paramagnetism is exhibited by molecules

- (a) Not attracted in a magnetic field
- (b) Containing only paired electrons
- (c) Carrying a positive charge
- (d) Carrying unpaired electrons**

Answer: (d) Carrying unpaired electrons

Explanation: The correct choice is Carrying unpaired electrons. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.



100. The higher number of unpaired electrons are in

- (a) Fe
- (b) Fe⁺
- (c) Fe⁺²
- (d) Fe⁺³

Answer: (d) Fe⁺³

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d⁰ and d¹⁰ species are diamagnetic.

